

Budget-Additive Pandora’s Problem: Corrected Partial Progress

Prepared as a rigorous repair note

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Abstract

Berger, Ezra, Feldman, and Fusco asked whether Pandora’s problem with budget-additive inspection costs admits a polynomial-time exact or approximate algorithm. This note does not resolve that open problem. It gives corrected partial progress. First, an oracle-given budget-additive cost function is learned exactly with $n + 1$ cost queries. Second, the problem is exactly solvable in polynomial time for every fixed number K of distinct value distributions, even when the budget-additive weights are arbitrary. The key canonicalization is that within one value-distribution class, a strategy may be assumed to open lower-weight boxes before higher-weight boxes. Third, a corrected rounding/abstraction lemma is proved using conditional sampling from explicit couplings; this repairs the false deterministic decision-tree transfer claim in the earlier draft. Finally, simple obstructions show why these results do not by themselves imply a PTAS for arbitrary finite-support instances.

Status. The results are *partial progress*, not a solution of the general budget-additive problem. They solve a nontrivial special case exactly, namely a constant number of distinct value laws, and they give a conditional approximation theorem for instances admitting a strong finite-type abstraction. They do not show that arbitrary instances admit such an abstraction, and they do not settle the open problem from [1].

1 Model

There are n boxes. Box i has an independent nonnegative random value V_i drawn from a known finite-support distribution D_i . A policy sequentially chooses unopened boxes to inspect. After observing an inspected value, it may either continue or stop. If $S(\pi)$ is the random set inspected by policy π , its utility in instance I is

$$U_I(\pi) = \mathbb{E} \left[\max_{i \in S(\pi)} V_i \right] - \mathbb{E}[c(S(\pi))],$$

where the maximum over the empty set is 0. The optimal value is denoted

$$\text{OPT}(I) = \sup_{\pi} U_I(\pi).$$

A cost function is *budget-additive*, or *capped-additive*, if

$$c(S) = \min \left\{ B, \sum_{i \in S} w_i \right\}, \quad B, w_1, \dots, w_n \geq 0. \quad (1)$$

The algorithm initially has explicit access to the value distributions and cost-query access to c : it may query $c(S)$ for chosen subsets $S \subseteq [n]$.

2 Exact reconstruction of the budget-additive cost

The oracle aspect of the budget-additive class is completely benign.

Lemma 2.1 (Exact reconstruction from $n + 1$ cost queries). *Let c be any function of the form (1). Query the cost oracle on $[n]$ and on each singleton, and set*

$$B^* := c([n]), \quad a_i := c(\{i\}) \quad (i = 1, \dots, n).$$

Then, for every $S \subseteq [n]$,

$$c(S) = \min \left\{ B^*, \sum_{i \in S} a_i \right\}. \quad (2)$$

Thus the oracle can be replaced exactly by the explicit representation $(B^, a_1, \dots, a_n)$ using $n + 1$ cost queries.*

Proof. Fix one representation $c(S) = \min\{B, \sum_{i \in S} w_i\}$. There are two cases.

First suppose $B \leq \sum_i w_i$. Then $B^* = c([n]) = B$, and $a_i = c(\{i\}) = \min\{B, w_i\}$. For every S ,

$$\min \left\{ B^*, \sum_{i \in S} a_i \right\} = \min \left\{ B, \sum_{i \in S} \min\{B, w_i\} \right\} = \min \left\{ B, \sum_{i \in S} w_i \right\} = c(S).$$

The middle equality holds because replacing any term $w_i > B$ by B cannot change the outer minimum with B .

Second suppose $B > \sum_i w_i$. Then the cap is never reached, so $B^* = c([n]) = \sum_i w_i$ and $a_i = w_i$ for every i . Since $\sum_{i \in S} a_i \leq B^*$ for every S ,

$$\min \left\{ B^*, \sum_{i \in S} a_i \right\} = \sum_{i \in S} w_i = c(S).$$

This proves (2). □

Remark 2.2. *The learned representation need not equal the original representation, but it is extensionally identical as a set function. From now on, after the $n + 1$ queries, we write the learned cost as*

$$c(S) = \min \left\{ B, \sum_{i \in S} a_i \right\}.$$

3 Exact dynamic programming for few value distributions

The earlier finite-alphabet idea grouped boxes by both value law and weight. For budget-additive costs, the weight grouping can be removed. If boxes have the same value distribution, then one may always open the cheaper boxes of that value class first.

Fix a partition of the boxes into value-distribution classes $\mathcal{T}$: boxes in the same class t have common distribution D_t . For class t , sort the learned weights increasingly,

$$a_{t,1} \leq a_{t,2} \leq \dots \leq a_{t,n_t}.$$

A policy is called *canonical* if, whenever it opens a box from class t after already opening j boxes from that class, it opens the box of weight $a_{t,j+1}$.

Lemma 3.1 (Cheapest-first canonicalization). *For every adaptive policy π , there is a canonical adaptive policy π^{can} such that*

$$U_I(\pi^{\text{can}}) \geq U_I(\pi).$$

Consequently, some optimal policy is canonical.

Proof. The canonical policy simulates π on virtual labels. In each class t , it keeps the count j_t of actual boxes already opened from that class. Whenever the simulated policy asks to open some unopened virtual box in class t , the canonical policy instead opens the cheapest actual unopened box in class t , namely the box of weight a_{t,j_t+1} . It feeds the observed value back to the simulation as the value of the virtual box requested by π .

Because boxes in the same class are independent and identically distributed, the sequence of values fed to the simulation has exactly the same joint law as the sequence of values that π would observe from the virtual labels. Hence the terminal maximum in the simulation has the same distribution as under π .

On each sample path, suppose the virtual policy has opened j_t boxes from each class t . The canonical policy has opened exactly the j_t cheapest boxes from each class. Therefore its total uncapped weight is no larger than that of the virtual opened set. Since $z \mapsto \min\{B, z\}$ is nondecreasing, its capped cost is also no larger path by path. Thus the expected reward is unchanged and the expected cost is weakly smaller. $\square$

Let $\mathcal{X} \subset \mathbb{R}_+$ be the union of the distribution supports, with 0 added if necessary. For each class t , define prefix sums

$$A_t(j) = \sum_{r=1}^j a_{t,r}, \quad j = 0, 1, \dots, n_t,$$

where $A_t(0) = 0$. For a count vector $j = (j_t)_{t \in \mathcal{T}}$, define the cost of the canonical opened set

$$C(j) = \min \left\{ B, \sum_{t \in \mathcal{T}} A_t(j_t) \right\}.$$

Let e_t denote the unit vector in coordinate t . The dynamic-programming state is (j, x) , where j_t is the number of already opened boxes from class t and $x \in \mathcal{X}$ is the current maximum observed value. Define

$$F(j, x) = \max \left\{ x, \max_{t: j_t < n_t} \left(- (C(j + e_t) - C(j)) + \sum_{y \in \mathcal{X}} D_t(y) F(j + e_t, \max\{x, y\}) \right) \right\}. \quad (3)$$

If $j_t = n_t$ for all t , the inner maximum is absent and $F(j, x) = x$.

Theorem 3.2 (Exact algorithm for K value-distribution types). *The recurrence (3) computes the optimal expected utility of the budget-additive Pandora instance. In particular,*

$$\text{OPT}(I) = F(0, 0),$$

where 0 denotes both the all-zero count vector and the initial maximum value. An optimal adaptive policy is obtained by choosing any action attaining the maximum in (3).

Let $K = |\mathcal{T}|$, $L = |\mathcal{X}|$, $s_t = |\text{supp}(D_t)|$, and

$$M = \prod_{t \in \mathcal{T}} (n_t + 1).$$

The table has ML states and can be filled in

$$O\left(ML \sum_{t \in \mathcal{T}} s_t\right) \subseteq O(MKL^2)$$

arithmetic operations and $O(ML)$ space. Hence, for fixed K , the problem is exactly solvable in polynomial time after the $n + 1$ reconstruction queries, assuming explicitly represented finite supports.

Proof. By Lemma 3.1, it suffices to optimize over canonical policies. Under a canonical policy, once the count vector is j , the opened set is determined up to irrelevant labels: it consists of the j_t cheapest boxes from each class t . Its capped cost is exactly $C(j)$.

At state (j, x) , the policy may stop and receive x . Otherwise it chooses a class t with $j_t < n_t$, opens the next cheapest box from that class, pays deterministic marginal cost $C(j + e_t) - C(j)$, observes $y \sim D_t$, and moves to $(j + e_t, \max\{x, y\})$. Conditional on (j, x) , no other information about the past affects the distribution of future values or costs. Bellman's principle of optimality gives (3).

The recurrence is acyclic because every opening increases $\sum_t j_t$ by one. Thus the table is filled by decreasing values of $\sum_t j_t$, starting from $\sum_t j_t = n$. There are ML states. For each state and each feasible class action t , evaluating the expectation requires summing over s_t support points. This proves the running-time and space bounds. $\square$

Corollary 3.3 (Identically distributed boxes). *If all boxes have the same value distribution with support size s , sorting the learned weights increasingly and applying (3) gives an exact optimal policy in $O(nLs)$ time after the $n + 1$ cost queries. In particular, this is $O(nL^2)$.*

Corollary 3.4 (Fixed number of value laws). *For every fixed K , budget-additive Pandora's problem with at most K distinct value distributions, arbitrary budget-additive weights, and explicit finite supports is exactly solvable in polynomial time and with $n + 1$ cost queries.*

4 Corrected policy transfer for rounded instances

The flawed transfer argument in the earlier draft tried to run an arbitrary deterministic decision tree directly on rounded observations. That is not valid: small perturbations may change the branch and hence the opened set. The correct transfer uses conditional sampling from couplings.

Definition 4.1 (Coupled abstraction). *Let I and $\tilde{I}$ be two Pandora instances on the same box labels, with value laws D_i and $\tilde{D}_i$ and costs c and $\tilde{c}$. We say that $\tilde{I}$ is an (η_V, η_C) -coupled abstraction of I if:*

- (i) *for every box i , there is a coupling γ_i of D_i and $\tilde{D}_i$ supported on pairs $(v, \tilde{v})$ satisfying $|v - \tilde{v}| \leq \eta_V$;*
- (ii) *for every set $S \subseteq [n]$, $|c(S) - \tilde{c}(S)| \leq \eta_C$.*

The couplings are regarded as part of the abstraction certificate.

Lemma 4.2 (Policy transfer through couplings). *If $\tilde{I}$ is an (η_V, η_C) -coupled abstraction of I , then*

$$|\text{OPT}(I) - \text{OPT}(\tilde{I})| \leq \eta_V + \eta_C.$$

Moreover, every policy for either instance can be simulated on the other, using the conditional kernels induced by the couplings, with expected utility loss at most $\eta_V + \eta_C$.

Proof. We prove the transfer from $\tilde{I}$ to I ; the other direction is symmetric. Let $\tilde{\pi}$ be a policy for $\tilde{I}$. Implement it on I as follows. Whenever $\tilde{\pi}$ asks to open box i , open box i in I , observe $V_i = v$, sample a pseudo-rounded value $\tilde{v}$ from the conditional distribution of $\tilde{V}_i$ given $V_i = v$ under γ_i , and feed $\tilde{v}$ to the internal simulation of $\tilde{\pi}$.

Equivalently, couple the execution by drawing independent pairs $(V_i, \tilde{V}_i) \sim \gamma_i$ as boxes are opened. The pseudo-rounded transcript has exactly the same law as a genuine execution of $\tilde{\pi}$ on $\tilde{I}$, including under adaptive selection, because the box pairs are independent. On every coupled sample path, the original execution and the pseudo-rounded execution open the same labeled set S , and $|V_i - \tilde{V}_i| \leq \eta_V$ for every opened i . Hence

$$\left| \max_{i \in S} V_i - \max_{i \in S} \tilde{V}_i \right| \leq \eta_V, \quad |c(S) - \tilde{c}(S)| \leq \eta_C.$$

The original pathwise utility is therefore at least the rounded pathwise utility minus $\eta_V + \eta_C$. Taking expectations proves the transfer. Reversing the conditional kernels proves the opposite transfer, and applying the two transfers to optimal policies gives the displayed optimal-value bound. $\square$

Theorem 4.3 (Conditional approximation from a few-value-law abstraction). *Fix $\varepsilon > 0$. Suppose that, after the $n + 1$ cost queries of Lemma 2.1, one constructs an (η_V, η_C) -coupled abstraction $\tilde{I}$ of I such that $\eta_V + \eta_C \leq \varepsilon/2$, and suppose $\tilde{I}$ has at most K distinct value laws. Solve $\tilde{I}$ exactly by Theorem 3.2 and lift the resulting policy to I using the conditional-kernel simulation in Lemma 4.2. The lifted policy π satisfies*

$$U_I(\pi) \geq \text{OPT}(I) - \varepsilon.$$

Its running time is the dynamic-programming time of Theorem 3.2 on the abstracted instance, plus the time needed to construct the abstraction.

Proof. Let $\tilde{\pi}^*$ be an optimal policy for $\tilde{I}$, and let π be its lifted implementation on I . By Lemma 4.2,

$$U_I(\pi) \geq U_{\tilde{I}}(\tilde{\pi}^*) - (\eta_V + \eta_C) = \text{OPT}(\tilde{I}) - (\eta_V + \eta_C).$$

The same lemma gives $\text{OPT}(\tilde{I}) \geq \text{OPT}(I) - (\eta_V + \eta_C)$. Combining the inequalities yields

$$U_I(\pi) \geq \text{OPT}(I) - 2(\eta_V + \eta_C) \geq \text{OPT}(I) - \varepsilon.$$

$\square$

5 Limitations and non-results

The preceding theorem is conditional. The required small abstraction does not automatically exist for arbitrary finite-support inputs.

Proposition 5.1 (Direct deterministic transfer is false). *It is false that an arbitrary deterministic decision tree can be run on rounded observations with utility loss bounded by the value-rounding error, even with zero costs and deterministic values.*

Proof. Take two boxes and zero costs. In the original instance, let

$$V_1 \equiv \frac{1}{2}, \quad V_2 \equiv 1.$$

In the rounded instance, let

$$\tilde{V}_1 \equiv \frac{1}{2} + \delta, \quad \tilde{V}_2 \equiv 1,$$

where $0 < \delta < 1/2$. The values are pointwise δ -close. Consider the deterministic strategy that opens box 1, then opens box 2 if and only if the observed value is at most $1/2$. On the original observations it opens box 2 and obtains utility 1. On the rounded observations the same tree stops after box 1 and obtains $1/2 + \delta$. The loss is $1/2 - \delta$, much larger than δ . $\square$

Proposition 5.2 (Many value laws may be necessary). *Fix $\eta_V < 1/2$. For every n , there is an n -box instance with values in $\{0, 1\}$ and zero costs such that any coupled abstraction satisfying the value condition in Definition 4.1 and assigning equal rounded laws to boxes of the same type requires n distinct rounded value laws.*

Proof. Let $V_i \sim \text{Bernoulli}(p_i)$ with $p_1, \dots, p_n$ pairwise distinct. If V_i is coupled to $\tilde{V}_i$ with $|V_i - \tilde{V}_i| \leq \eta_V < 1/2$, then the intervals $[-\eta_V, \eta_V]$ and $[1 - \eta_V, 1 + \eta_V]$ are disjoint and

$$\Pr[\tilde{V}_i \in [1 - \eta_V, 1 + \eta_V]] = \Pr[V_i = 1] = p_i.$$

Thus the rounded law determines p_i . Two boxes with different Bernoulli parameters cannot share the same rounded law. $\square$

Proposition 5.3 (Uniform subset-cost rounding controls total weight error). *Consider the additive special case $c(S) = \sum_{i \in S} a_i$, which is budget-additive with a nonbinding cap. If another additive cost $\tilde{c}(S) = \sum_{i \in S} \tilde{a}_i$ satisfies*

$$\sup_{S \subseteq [n]} |c(S) - \tilde{c}(S)| \leq \eta_C,$$

then

$$\sum_{i=1}^n |a_i - \tilde{a}_i| \leq 2\eta_C.$$

Proof. Let $e_i = a_i - \tilde{a}_i$. Taking $S_+ = \{i : e_i > 0\}$ gives

$$\sum_{i: e_i > 0} e_i = c(S_+) - \tilde{c}(S_+) \leq \eta_C.$$

Taking $S_- = \{i : e_i < 0\}$ gives

$$\sum_{i: e_i < 0} (-e_i) = \tilde{c}(S_-) - c(S_-) \leq \eta_C.$$

Adding the two inequalities proves the claim. $\square$

Remark 5.4. Proposition 5.2 rules out the claim that the few-value-law dynamic program automatically gives a PTAS for arbitrary finite-support instances. Proposition 5.3 shows that the uniform set-cost guarantee in the abstraction theorem is also a strong global requirement; naive independent rounding of singleton weights is not enough.

6 Conclusion

The repaired results are genuine but limited. Budget-additive costs can be learned exactly from $n + 1$ cost queries. Instances with a fixed number of distinct value distributions can then be solved exactly in polynomial time, even with arbitrary budget-additive weights. A valid approximation theorem follows from a certified coupled abstraction with few value laws. These statements are partial progress on the budget-additive question of Berger, Ezra, Feldman, and Fusco; they do not solve the general case with arbitrarily many distinct value distributions.

References

- [1] Ben Berger, Tomer Ezra, Michal Feldman, and Federico Fusco. Pandora’s Problem with Combinatorial Cost. *Mathematics of Operations Research*, 50(4):3161–3189, 2025. Published online 2024; conference version in *EC 2023*. DOI: [10.1287/moor.2023.0248](https://doi.org/10.1287/moor.2023.0248).
- [2] Martin L. Weitzman. Optimal search for the best alternative. *Econometrica*, 47(3):641–654, 1979.